

Latest Trends in Biomedical Instrumentation

Introduction

- **Biomedical Instrumentation = medical devices + electronics + sensors**
- Used for diagnosis, monitoring, therapy
- Rapid evolution due to:
 - AI & Data Science
 - IoT
 - Miniaturization
- **Focus:** Smart, real-time, patient-centric healthcare

Need for Advanced Instrumentation

- Rising chronic diseases
- Remote healthcare demand
- Precision medicine
- Real-time monitoring necessity
- Cost-effective healthcare solutions

Classification of Biomedical Instruments

- Diagnostic (ECG, MRI)
- Monitoring (ICU monitors)
- Therapeutic (Pacemakers)
- Imaging systems
- Wearable devices

Trend 1 – Wearable Health Devices

- Smart watches, fitness bands
- Continuous monitoring:
 - Heart rate
 - SpO₂
 - Sleep patterns
- Examples:
 - Apple Watch Series 9
 - Fitbit Charge 6
- Advantage: Preventive healthcare

Wearable Device Working

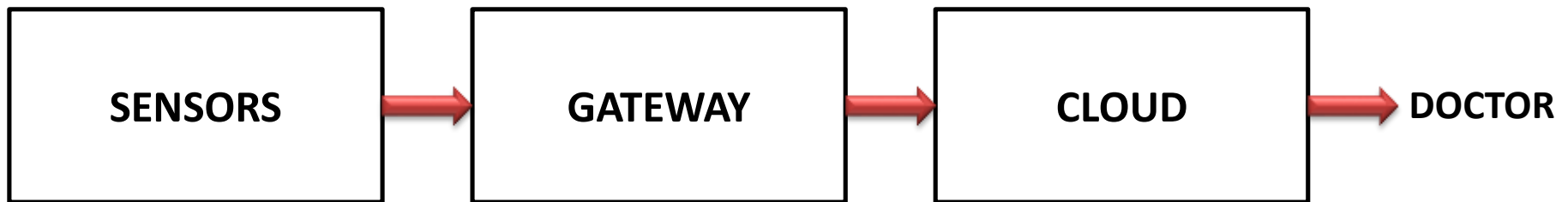
- Sensors → Signal acquisition
- Signal conditioning → Filtering
- ADC → Digital conversion
- Microcontroller → Processing
- Wireless → Data transmission

Trend 2 – Internet of Medical Things (IoMT)

- Connected medical devices via internet
- Real-time data sharing
- Remote diagnosis
- **Applications:**
 - Smart ICU
 - Remote patient monitoring
 - Home healthcare

IoMT Architecture

- Data analytics
- Alerts & decision making



Slide 9: Trend 3 – Artificial Intelligence (AI)

- AI in diagnostics
- Pattern recognition
- Disease prediction
- **Examples:**
 - Cancer detection
 - ECG analysis
 - Radiology imaging

AI in ECG Analysis

- Automated arrhythmia detection
- Faster diagnosis
- Reduced human error

Trend 4 – Telemedicine

- Remote consultation
- Video conferencing
- Digital prescriptions
- **Benefits:**
 - Rural healthcare access
 - Time & cost saving

Trend 5 – Point of Care (POC) Devices

- Portable diagnostic devices
- Immediate results
- **Examples:**
 - Glucose monitor
 - Portable ECG

Trend 6 – Implantable Devices

- Pacemakers
- Neurostimulators
- Cochlear implants
- Long-term monitoring & therapy

Trend 7 – Robotics in Surgery

- Precision surgery
- Minimally invasive
- **Example:**
 - Da Vinci Surgical System

Slide 15: Trend 8 – Biosensors

- Detect biological parameters
- High sensitivity
- **Applications:**
 - Glucose monitoring
 - DNA detection

Trend 9 – Lab-on-Chip Technology

- Miniaturized labs
- Fast diagnostics
- Low sample requirement

Challenges

- Data security (Cyber threats)
- High cost
- Regulatory approvals
- Accuracy & reliability

Future Scope

- Smart hospitals
- AI-driven diagnosis
- Personalized medicine
- Fully automated healthcare systems

A smart ventilator measures airway pressure using a sensor with output **0–5 V** for **0–50 cmH₂O**. The signal is digitized using a **10-bit ADC (0–5 V)**.

Find:

1. ADC resolution in mV/bit
2. Pressure resolution in cmH₂O/bit
3. Whether the system can detect a pressure change of **0.5 cmH₂O**

Solution:

ADC resolution:

$$\text{Resolution} = \frac{5}{2^{10}} = \frac{5}{1024} = 4.88 \text{ mV/bit}$$

Pressure sensor sensitivity:

$$\frac{50 \text{ cmH}_2\text{O}}{5 \text{ V}} = 10 \text{ cmH}_2\text{O/V}$$

Pressure resolution:

$$\begin{aligned} &4.88 \text{ mV} \times 10 \text{ cmH}_2\text{O/V} \\ &= 4.88 \times 10^{-3} \times 10 = 0.0488 \text{ cmH}_2\text{O/bit} \end{aligned}$$

Answer:

- ADC resolution = 4.88 mV/bit
- Pressure resolution = 0.0488 cmH₂O/bit

Since $0.0488 < 0.5$, the system **can detect** a 0.5 cmH₂O change.

An infusion pump monitors drug delivery rate. Sensor output is **0–4 V** for **0–100 mL/hr**, connected to an **8-bit ADC (0–5 V)**.

Find:

1. ADC resolution
2. Flow-rate resolution in mL/hr per bit
3. Whether the system can detect a change of **1 mL/hr**

Solution:

ADC resolution:

$$\frac{5}{2^8} = \frac{5}{256} = 19.53 \text{ mV/bit}$$

Sensor sensitivity:

$$\frac{100}{4} = 25 \text{ mL/hr per V}$$

Flow resolution:

$$\begin{aligned} & 19.53 \text{ mV} \times 25 \text{ mL/hr per V} \\ &= 19.53 \times 10^{-3} \times 25 = 0.488 \text{ mL/hr per bit} \end{aligned}$$

Answer:

- ADC resolution = **19.53 mV/bit**
- Flow-rate resolution = **0.488 mL/hr/bit**

A portable ECG device samples the signal at **200 Hz**. A patient's heart rate is **120 bpm**.

Find:

1. ECG signal frequency
2. Time period
3. Samples per cycle
4. Whether the sampling is sufficient

Solution:

Heart rate frequency:

$$f = \frac{120}{60} = 2 \text{ Hz}$$

Time period:

$$T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ s}$$

Samples per cycle:

$$200 \times 0.5 = 100$$

Answer:

- Frequency = **2 Hz**
- Time period = **0.5 s**
- Samples per cycle = **100**

Since 100 samples are obtained per cycle, sampling is **more than sufficient**.

A medical imaging sensor produces **0–1 V output**, digitized using a **10-bit ADC (0–5 V)**.

Find:

1. ADC resolution
2. Effective number of ADC levels used for 0–1 V range
3. Suggest improvement

Solution:

ADC resolution:

$$\frac{5}{1024} = 4.88 \text{ mV/bit}$$

Number of levels used over 0–1 V:

$$\frac{1}{4.88 \times 10^{-3}} \approx 205$$

Answer:

- ADC resolution = 4.88 mV/bit
- Effective levels used = 205 levels approximately

Suggestion:

Only a small portion of full ADC range is used. Improvement:

- amplify the signal to 0–5 V, or
- use a lower input-range ADC.

